

# Predicting Breast Cancer Survival

Five-year Survival Prediction Of The Breast Cancer Patients From UK & Canada  
Based On The Clinical Information

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# Introduction

## Context & Clinical motivation:

- Breast cancer is among the leading cause of death
- Outcome of treatments become important for customized therapeutic and improving patient outcomes

## Questions:

- A. How does the survival differ between different breast cancer subtypes?
- B. Can we predict 5-year survival of the breast cancer patients given clinical information?



# Introduction

## Dataset:

- Patient demographics
- Pathology
- Tumor grades
- Receptor classifications

## Novelty:

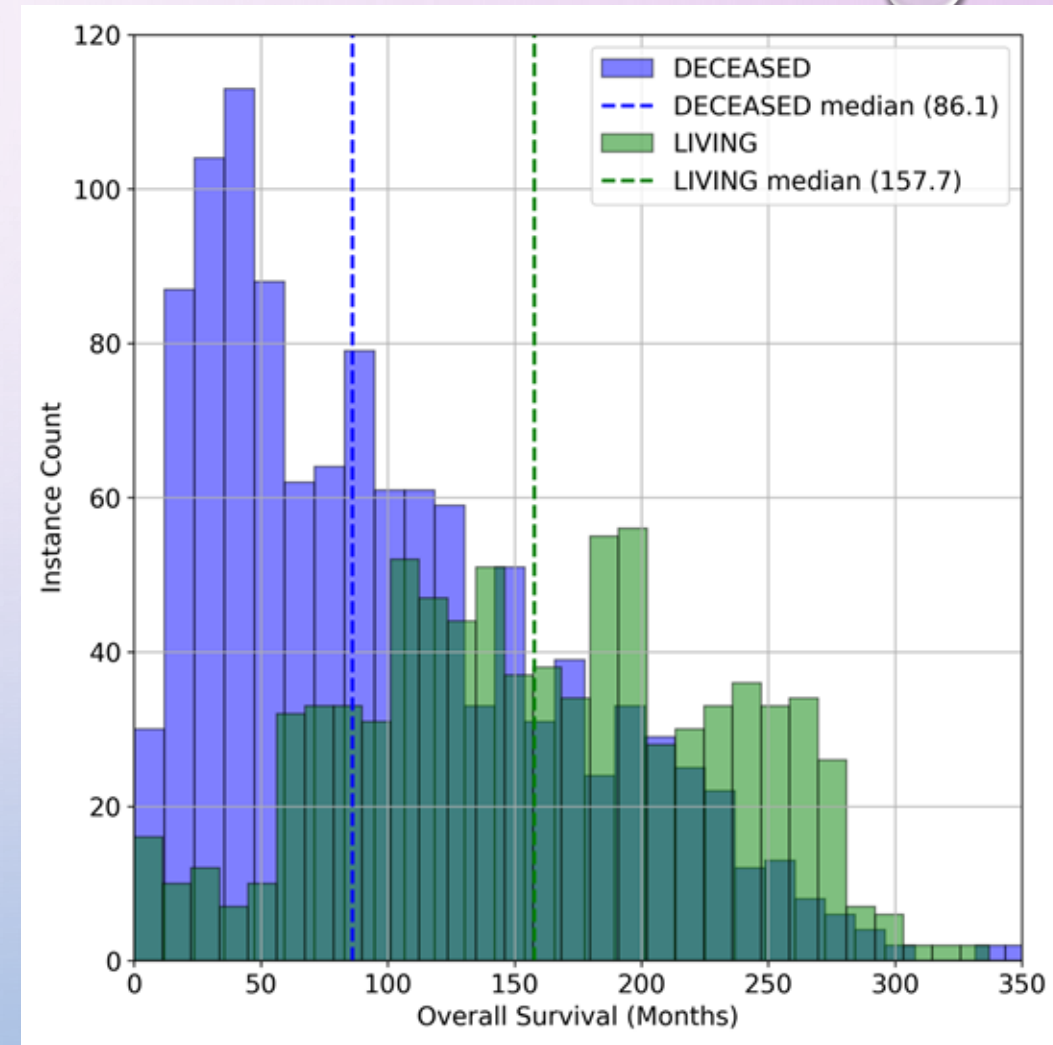
- Clinically relevant way of handling missing data
- Integrating genomic information to clinical features for model improvement
- Versatile model to predict survival of patients from different hospitals

## KPIs:

- Main KPIs: ROC-AUC, PR-AUC, MCC and F1
- Precision and Recall

## Stakeholders:

- Patients
- Researchers, Hospitals & Pharma companies
- Insurance companies



# Initial Data Audit

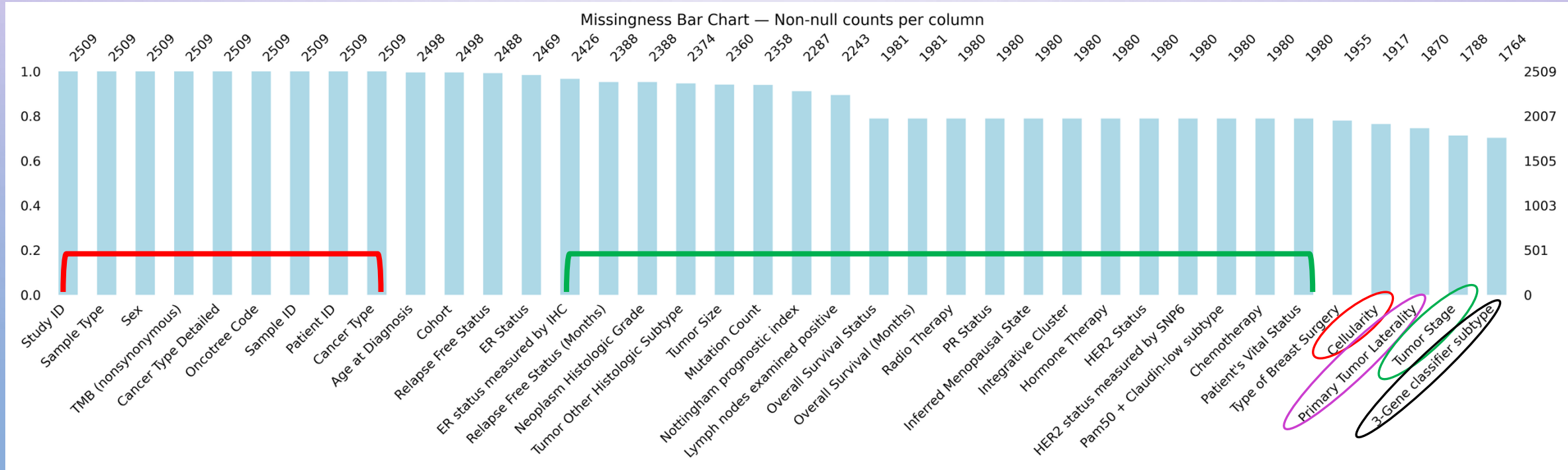
**Cohort Completeness Overview:** profiles data density across all clinical, and demographic features for 2,509 patients.

**100% Complete Fields:** Demographics: Study Id, Sex, & Age

**Clinical Variables:** Clinical and Therapy variables stabilize at exactly 1,980 non-null records (~79% density).

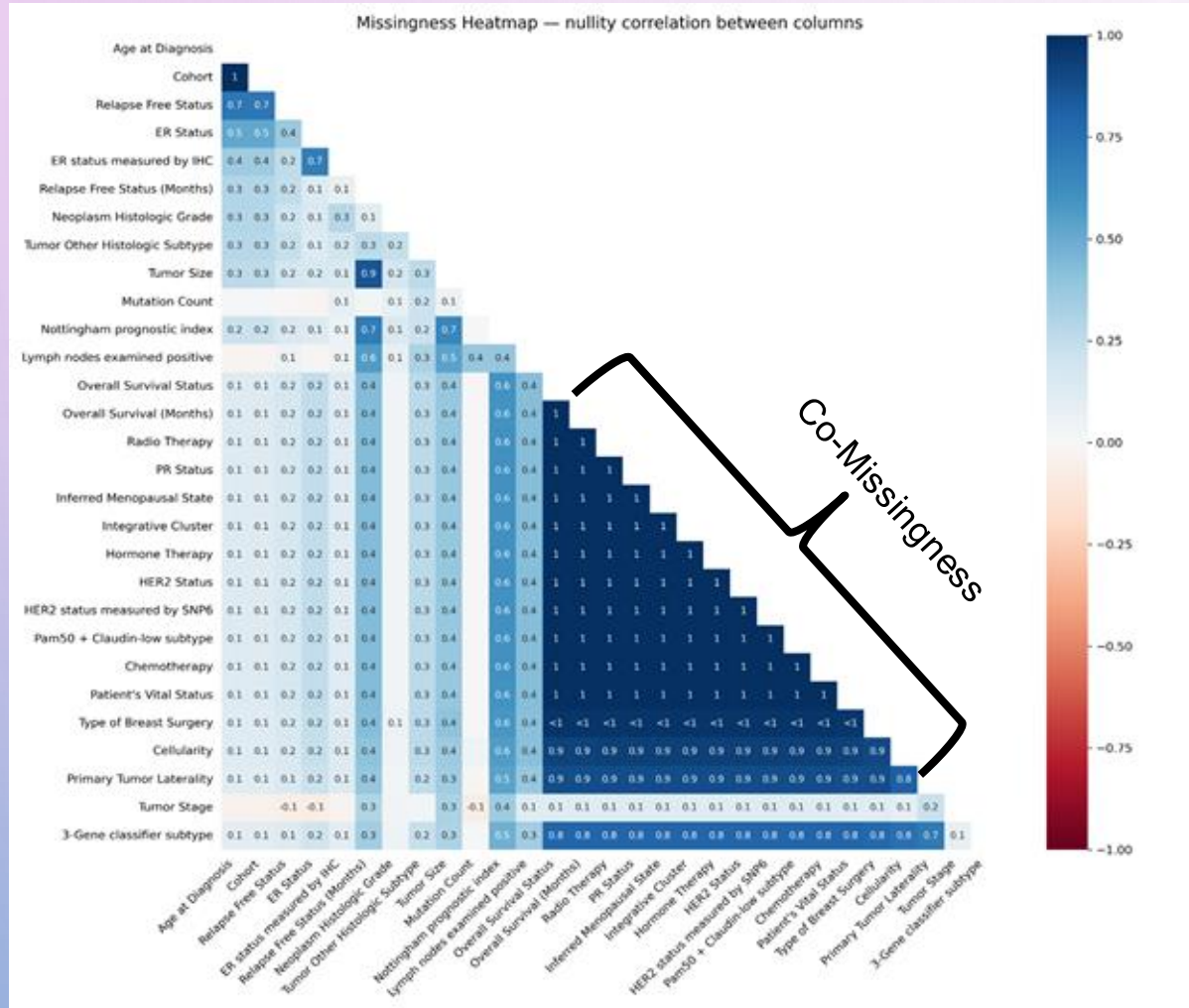
**Maximum Sparsity Risk:** Cellularity, Primary Tumor Laterality, Tumor Stage, and 3-Gene classifier subtype drop to 1,764 records (~70% dense).

**Imputation Justification:** High scattered missingness rules out row deletion

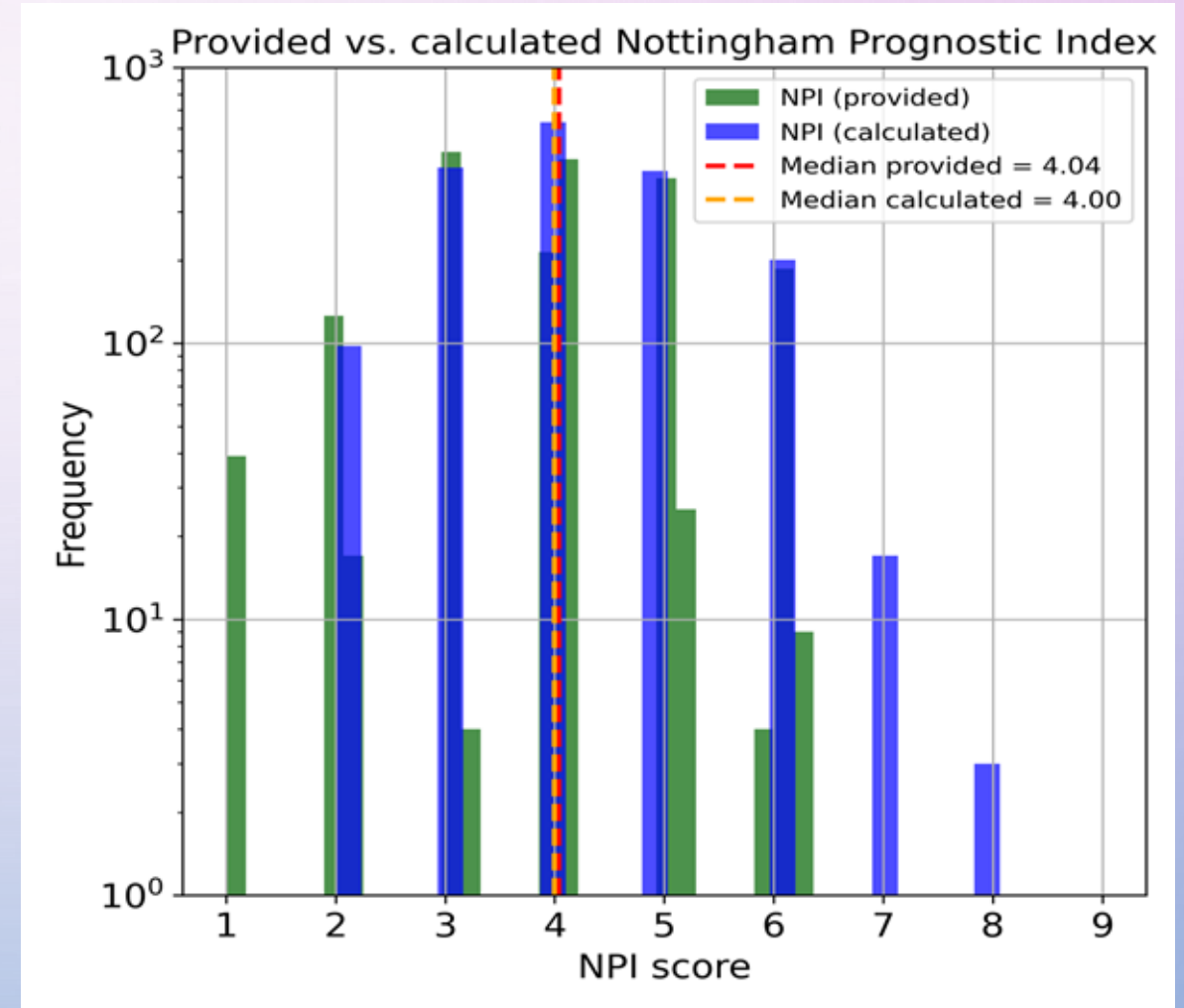




# Initial Data Audit



**Nullity Correlation:** Absence of data in features is correlated

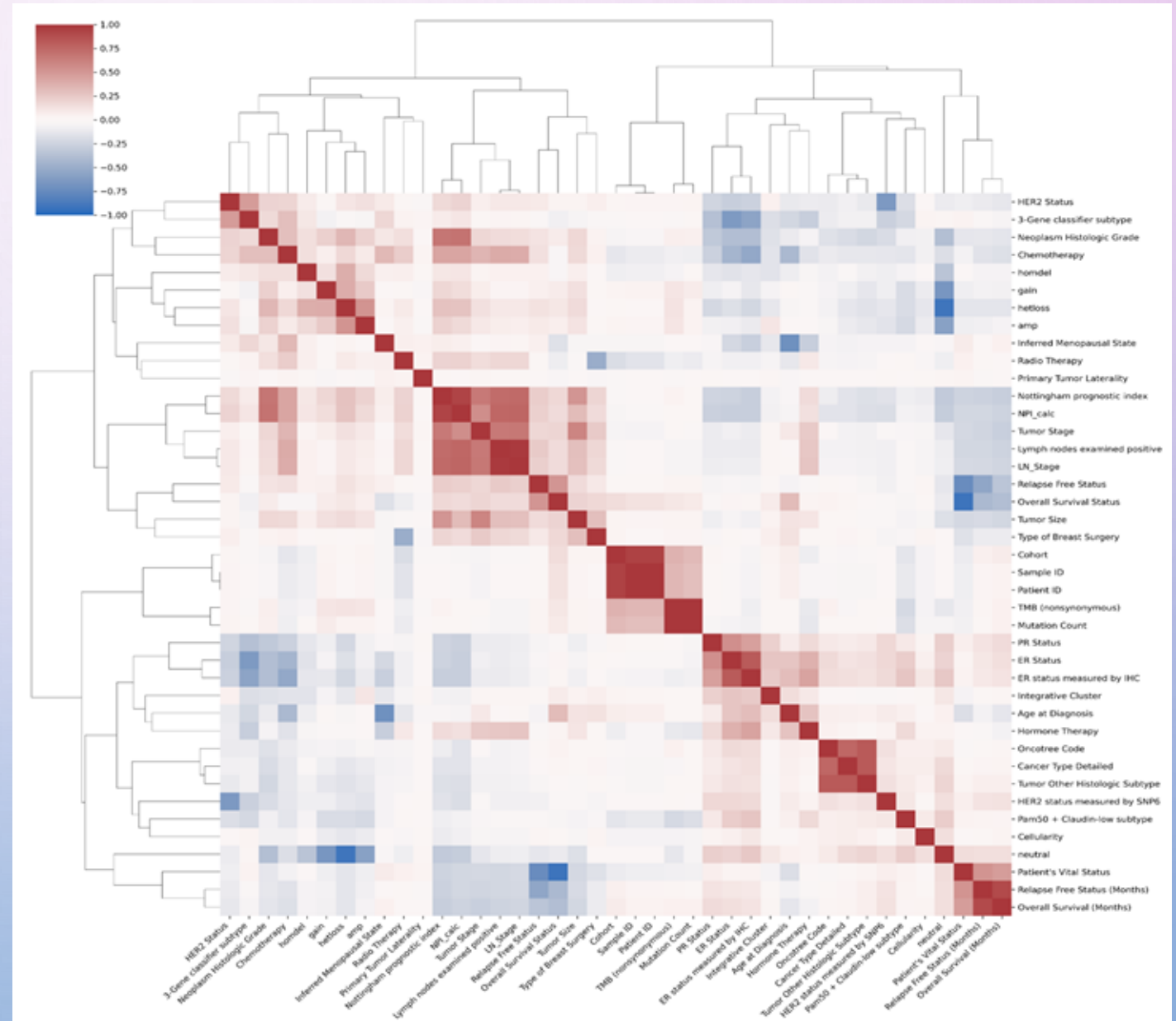


NPI  
 $= 0.2 \times \text{tumor diameter} + \text{histological grade (1-3)}$   
 $+ \text{lymph node stage (1-3)}$

# Feature Correlation & Clustering

**Multivariate Clusters:** Hierarchical clustering features into modules based on correlation, exposing biological and clinical relationships.

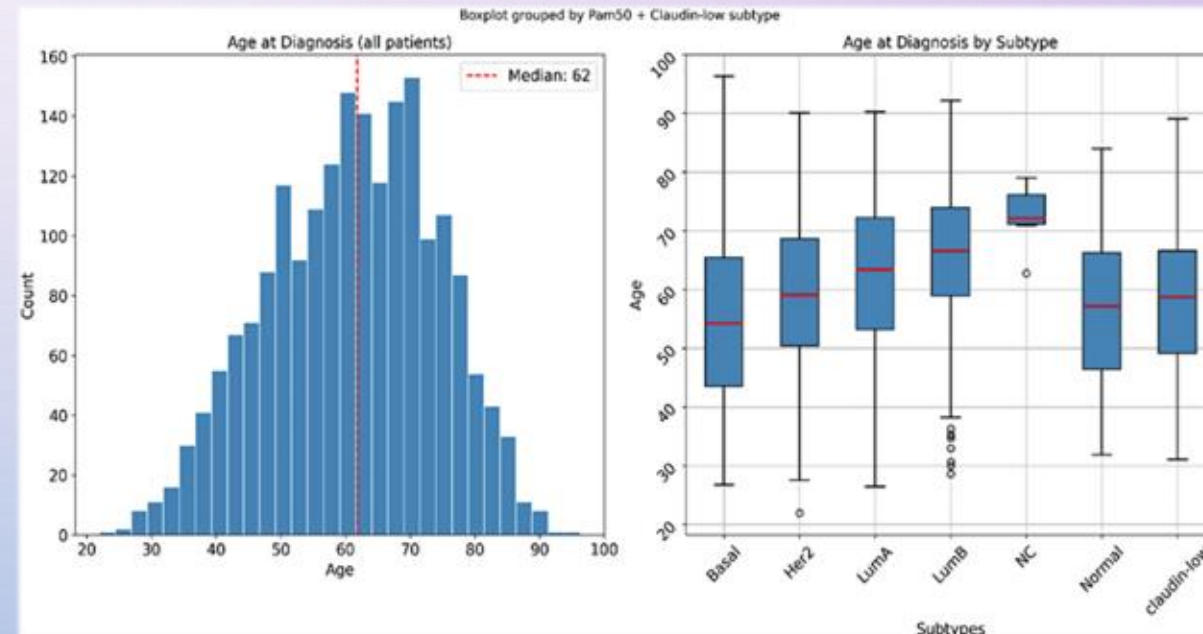
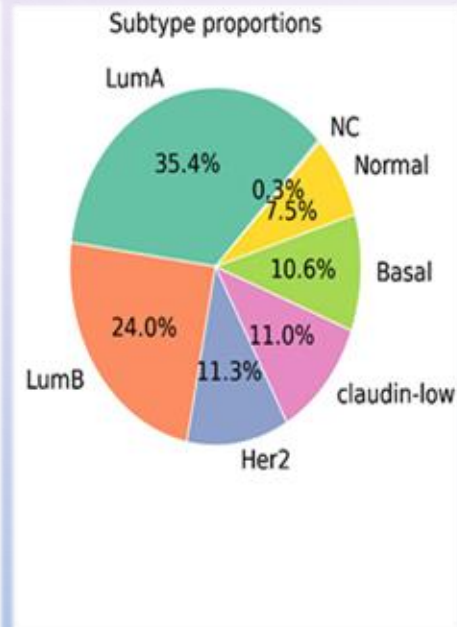
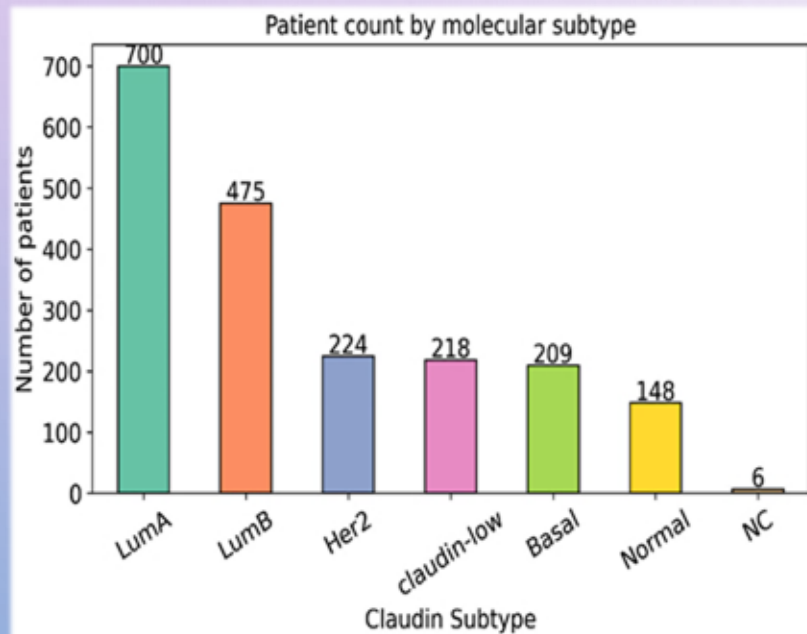
1. **Enzyme block:** PR Status, ER Status, ER status measured by IHC
2. **Cancer Type block:** Oncotree Code, Cancer Type Detailed, Tumor Other Histologic Subtype
3. **NPI block:** Nottingham prognostic index, NPI\_calc, LN\_Stage, Lymph nodes examined positive, Neoplasm Histologic Grade, Tumor Stage
4. **Mutation block:** Mutation Count, TMB (nonsynonymous)
5. **HER2, 3-gene block:** 3-Gene classifier subtype, HER2 Status
6. **CNA blocks:** homdel, hetloss, neutral, gain, amp



# COHORT COMPOSITION BY SUBTYPE

## Problem Statement 1:

How does survival differ across breast cancer subtypes — and how much worse off are triple-negative patients compared to other cancer subtypes?

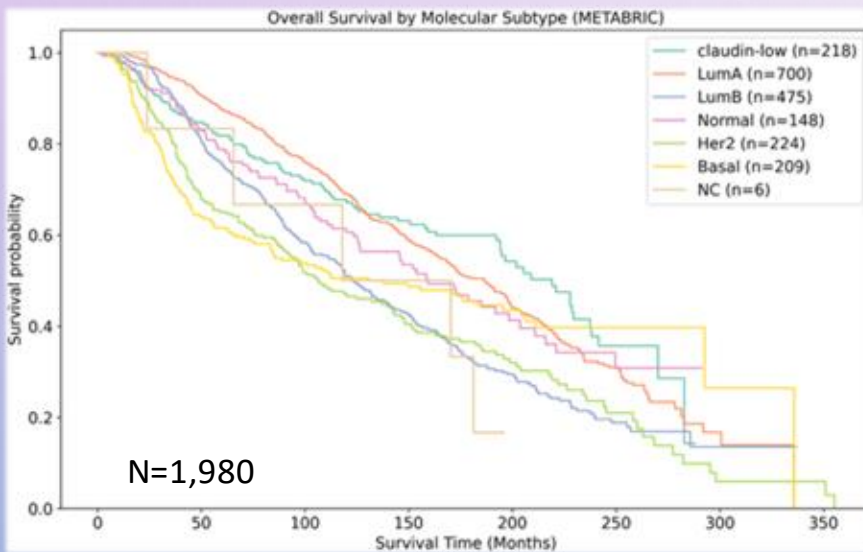


Subtype mix: LumA (~38%) & LumB (24%) dominate;  
Basal (10.6%) & Claudin-low (11%) are the TNBC-risk groups

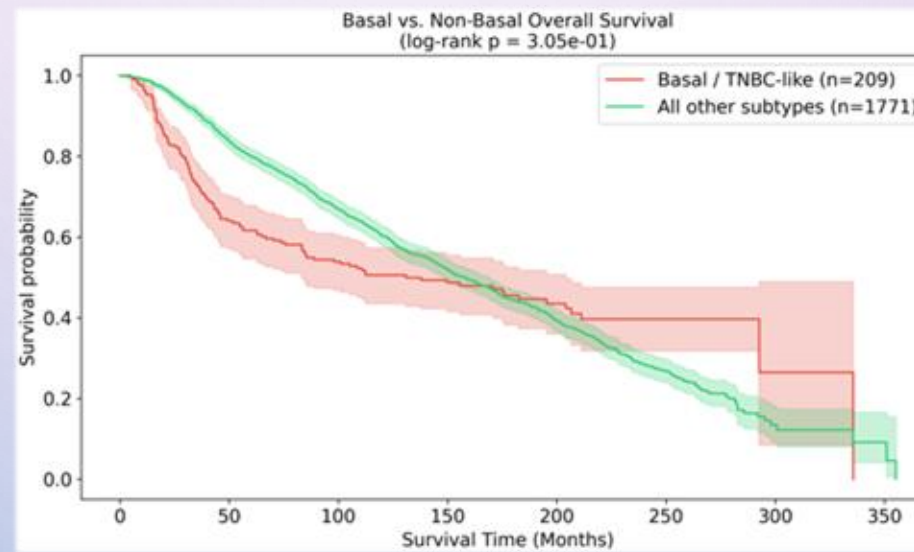
Age at diagnosis: median 62; Basal & Claudin-low skew  
younger — consistent with known TNBC biology

# SURVIVAL DIFFERS SHARPLY BY SUBTYPE

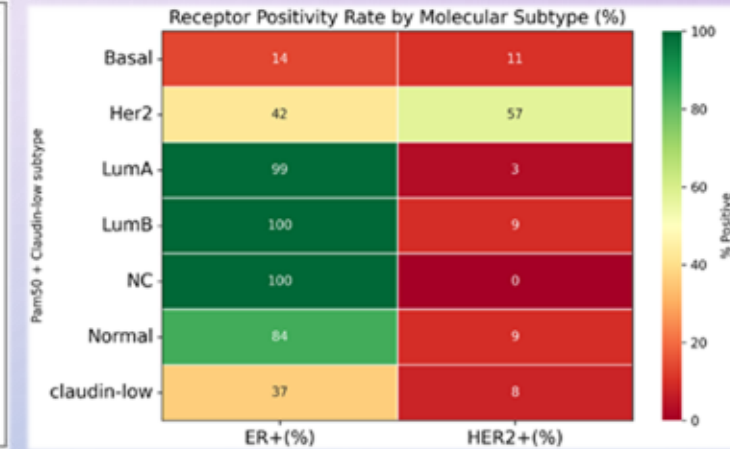
Basal/TNBC-like tumors have the poorest early survival — confirmed by the receptor profile



*KM curves diverge early —  
Basal shows the steepest decline of any subtype  
(log-rank  $p = 0$ ) across all 7 subtypes)*



*Receptor profile confirms Basal  $\approx$   
ER-/HER2*



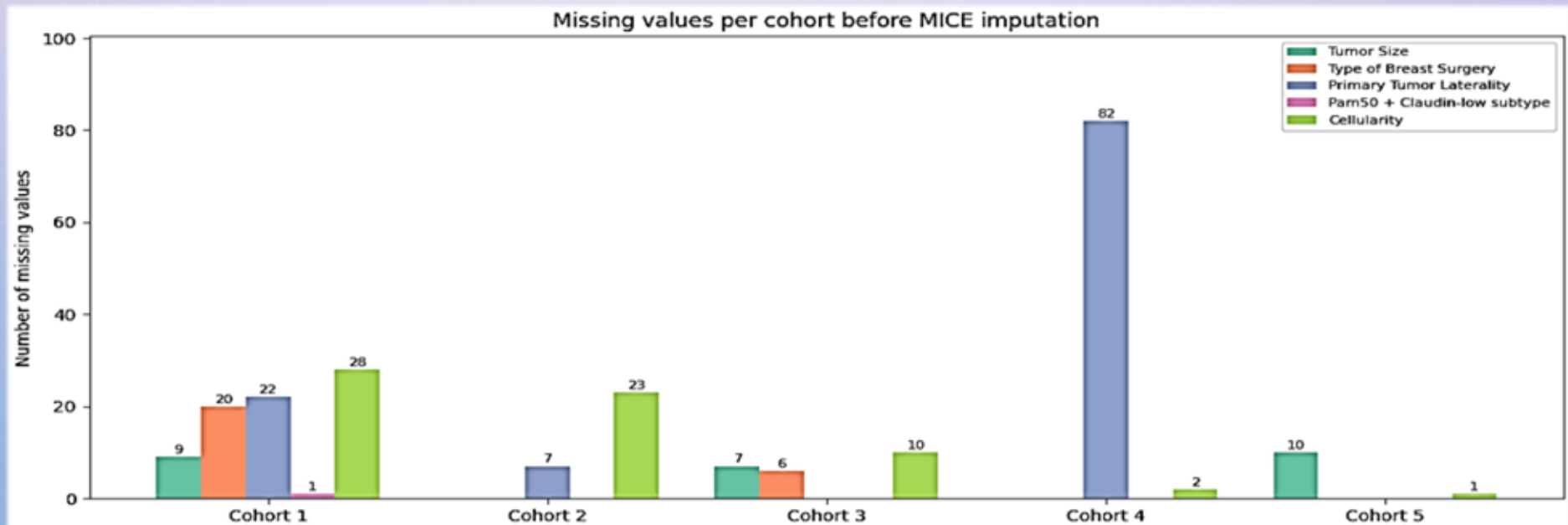
*Basal vs. rest confirms the gap  
(pairwise  $p = 0.31$ , not significant  
given small Basal  $n$ )*



# HANDLING MISSING DATA WITH MICE

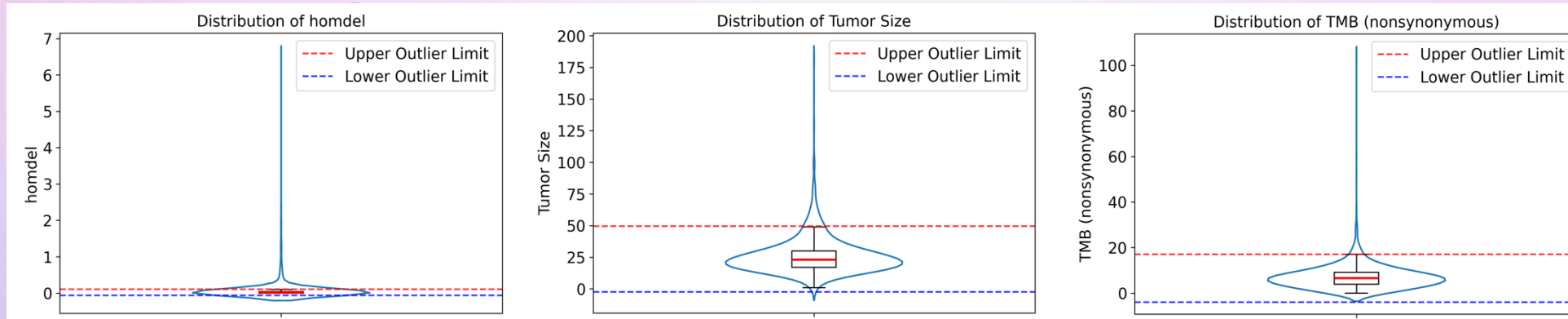
## MULTIVARIATE IMPUTATION BY CHAINED EQUATIONS

- **Why MICE:** Missingness was scattered across all selected features numeric/categorical columns (Tumor Size, Cellularity, Tumor Laterality, etc.) — too much data loss to simply drop rows.
- **How it works:** MICE models each feature from the others, capturing realistic joint relationships rather than filling with a single mean/mode.
- **Per-cohort:** Imputation runs separately within each of the 5 cohorts, so no cohort's data leaks into another's imputed values.

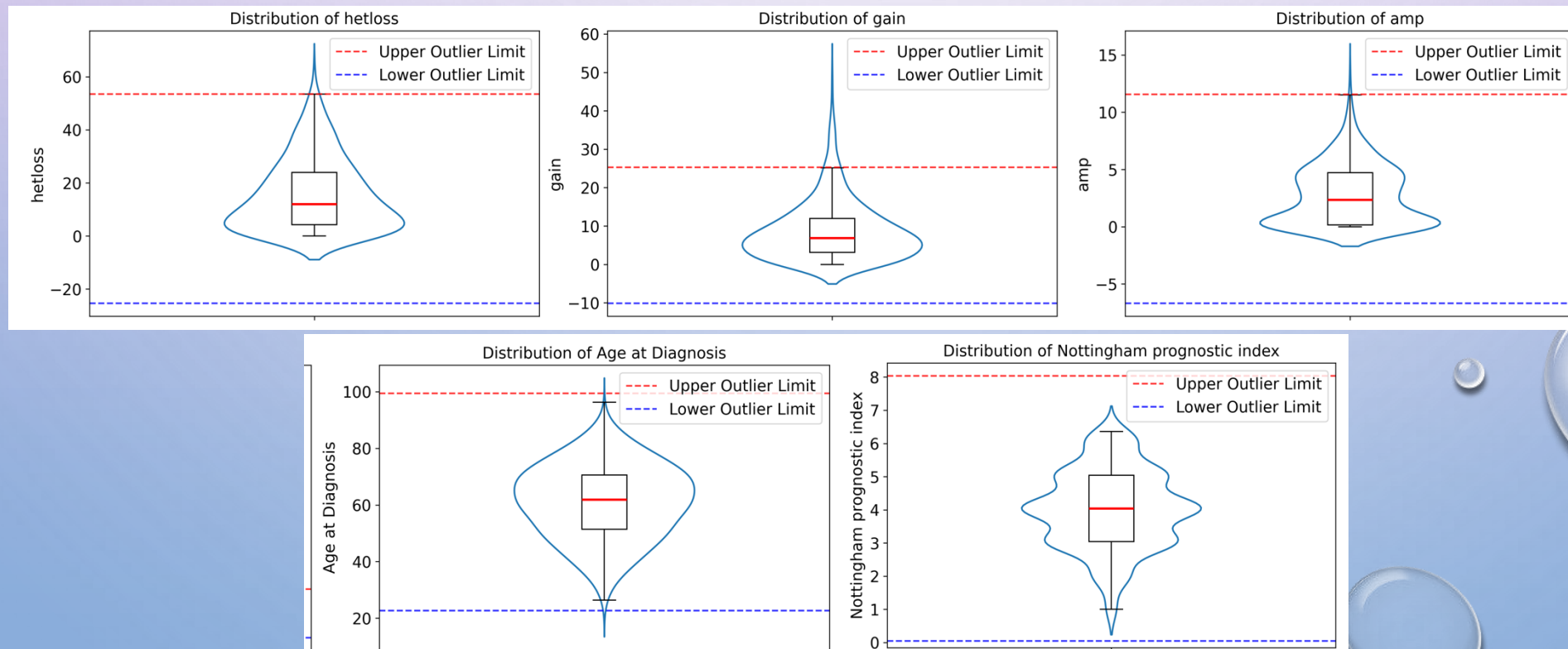


# Data Integrity & Feature Audit

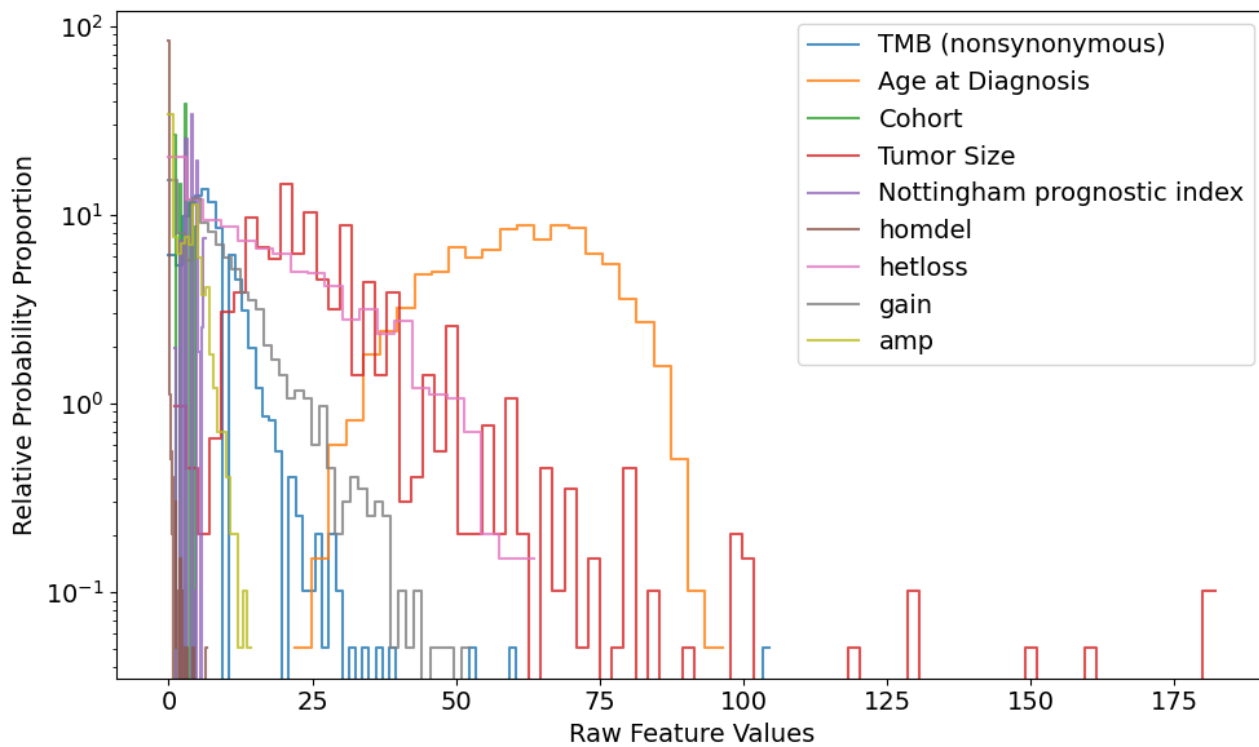
## Right Skew



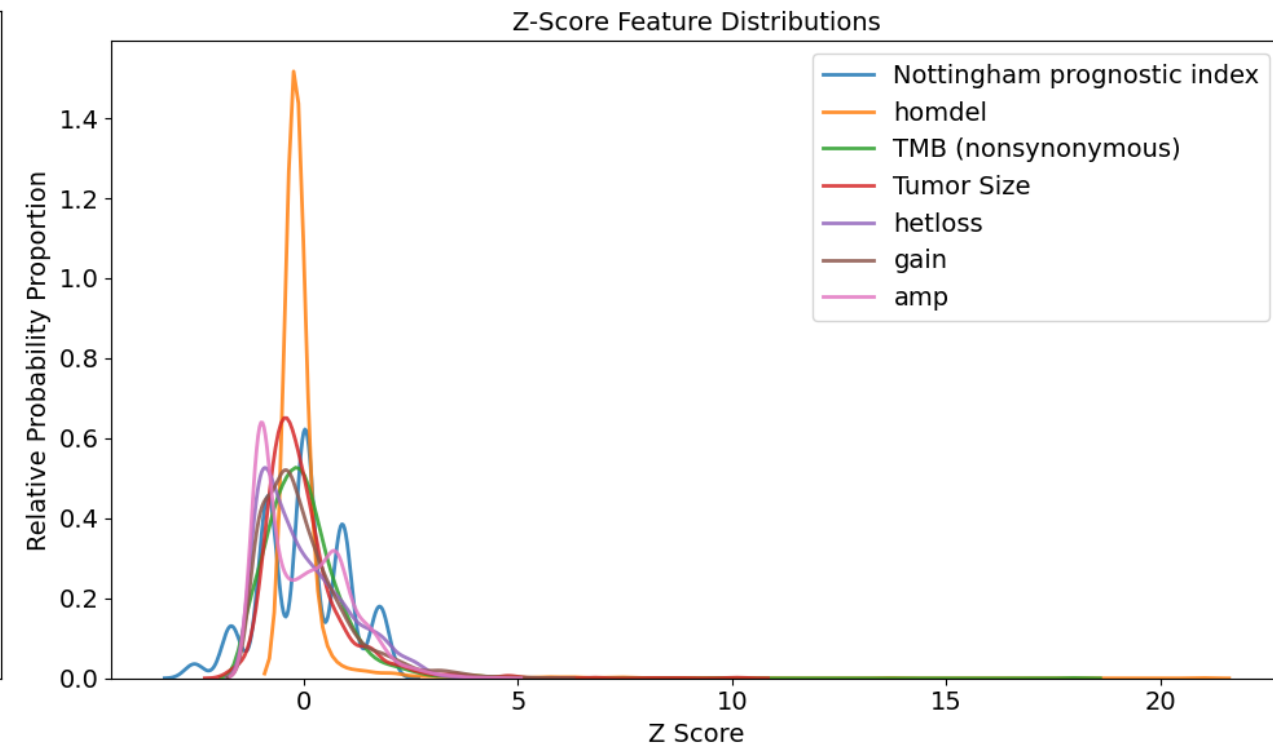
## Neutral Distributions



# Is Cohort A Good Choice For CV?



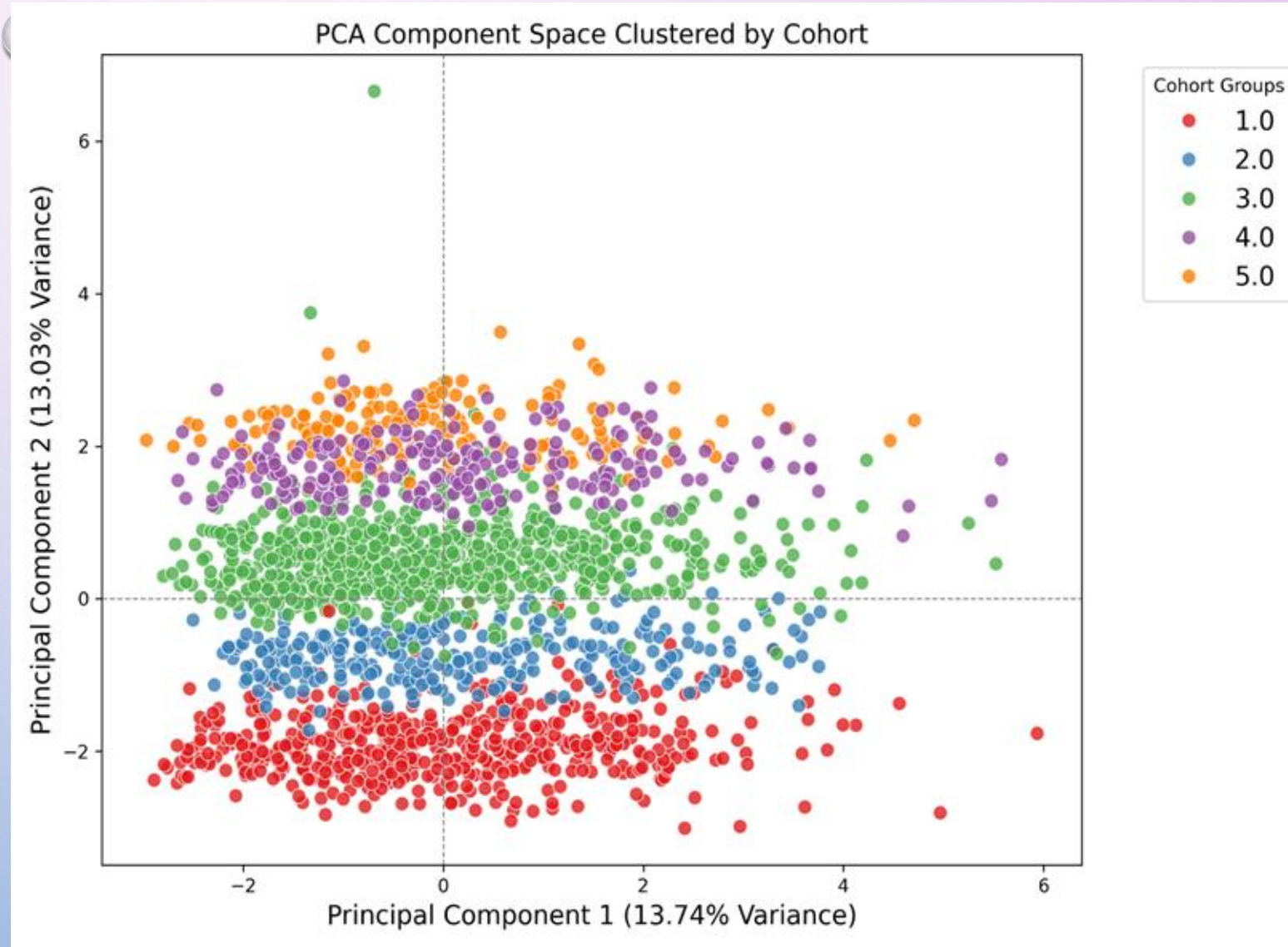
A



B

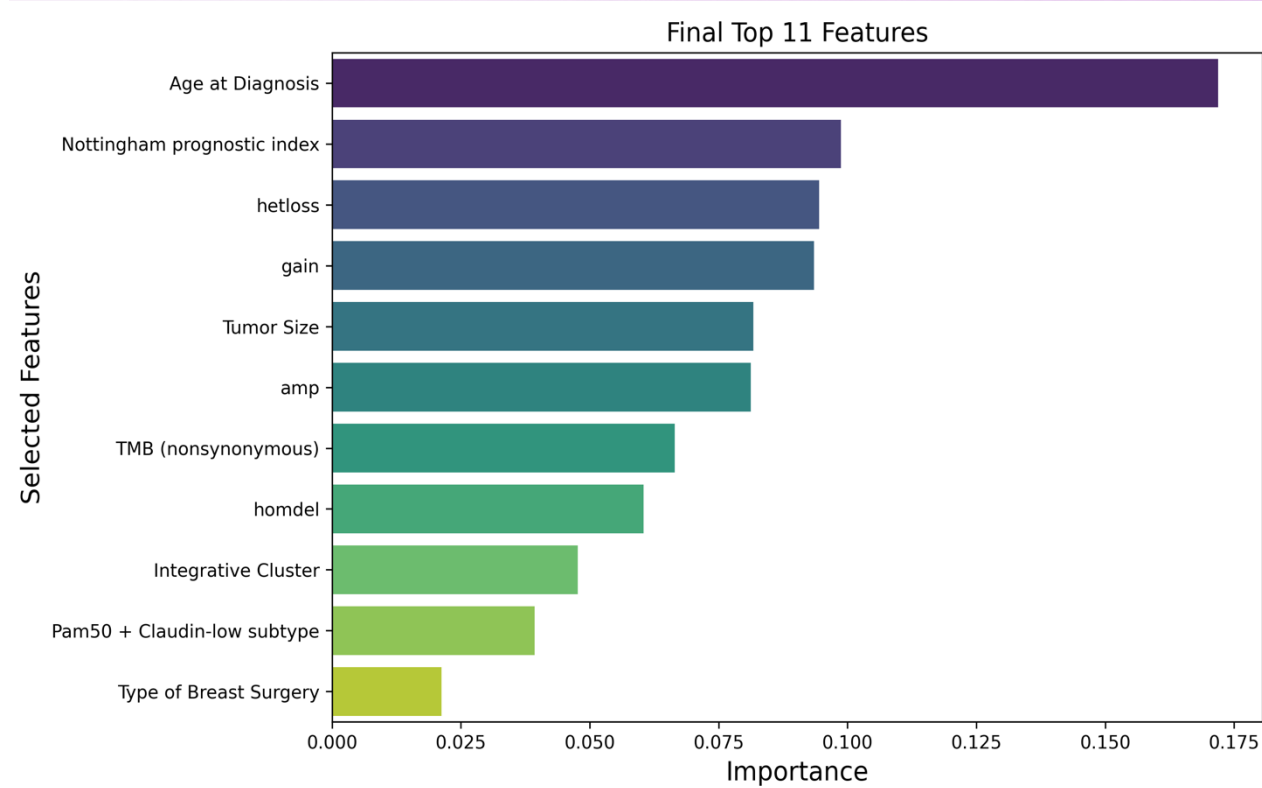
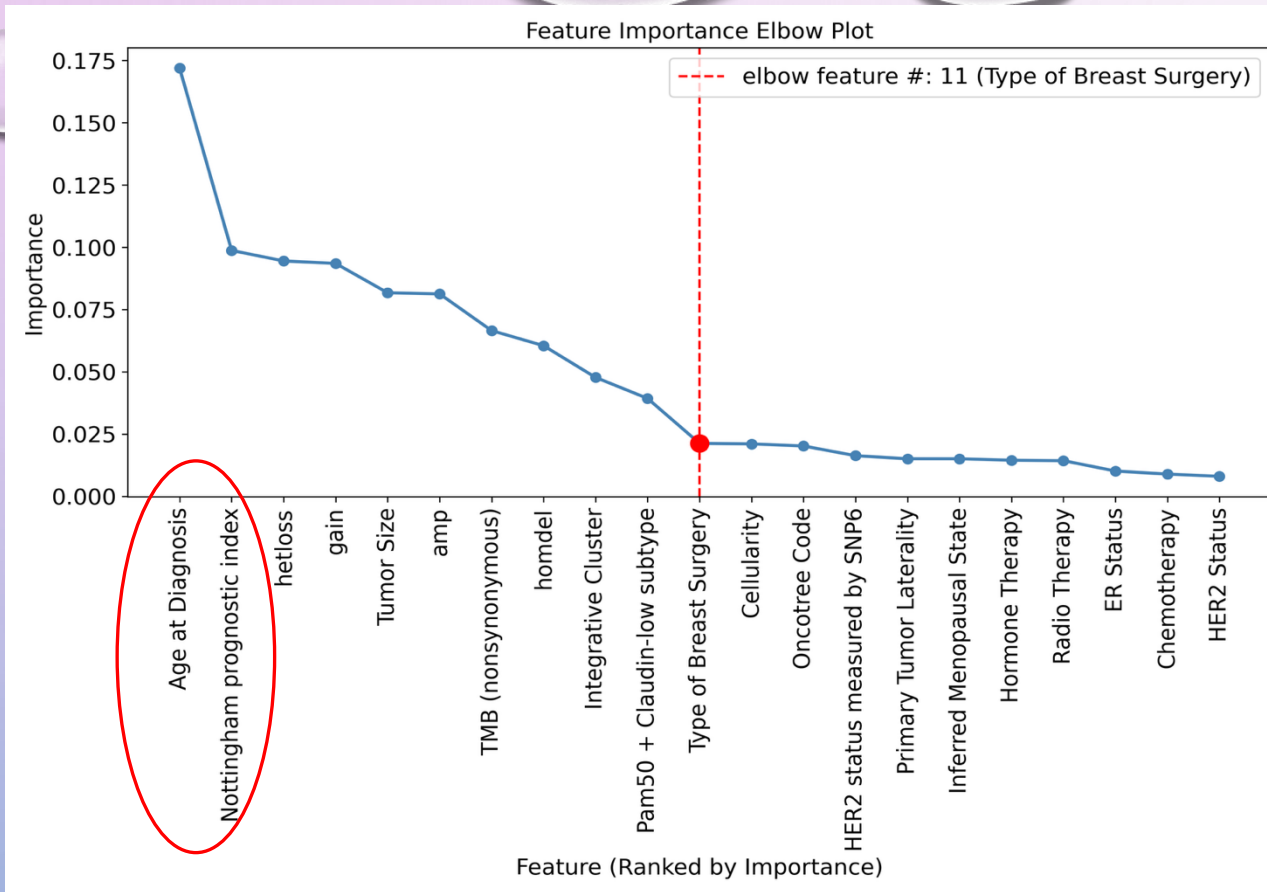
# Is Cohort A Good Choice For CV?

- Principal Component Analysis showed non-overlapping cohorts (1–5) along PC2
- Inter-cohort separation refutes **random splitting**
- Utilized **cohort-based cross-validation** to maintain structural integrity.





# RF-based Feature Selection & Elbow Optimization



**Elbow Method Selection:** Identifies **inflection point (red)** of importance, optimizing our feature space.

A

B

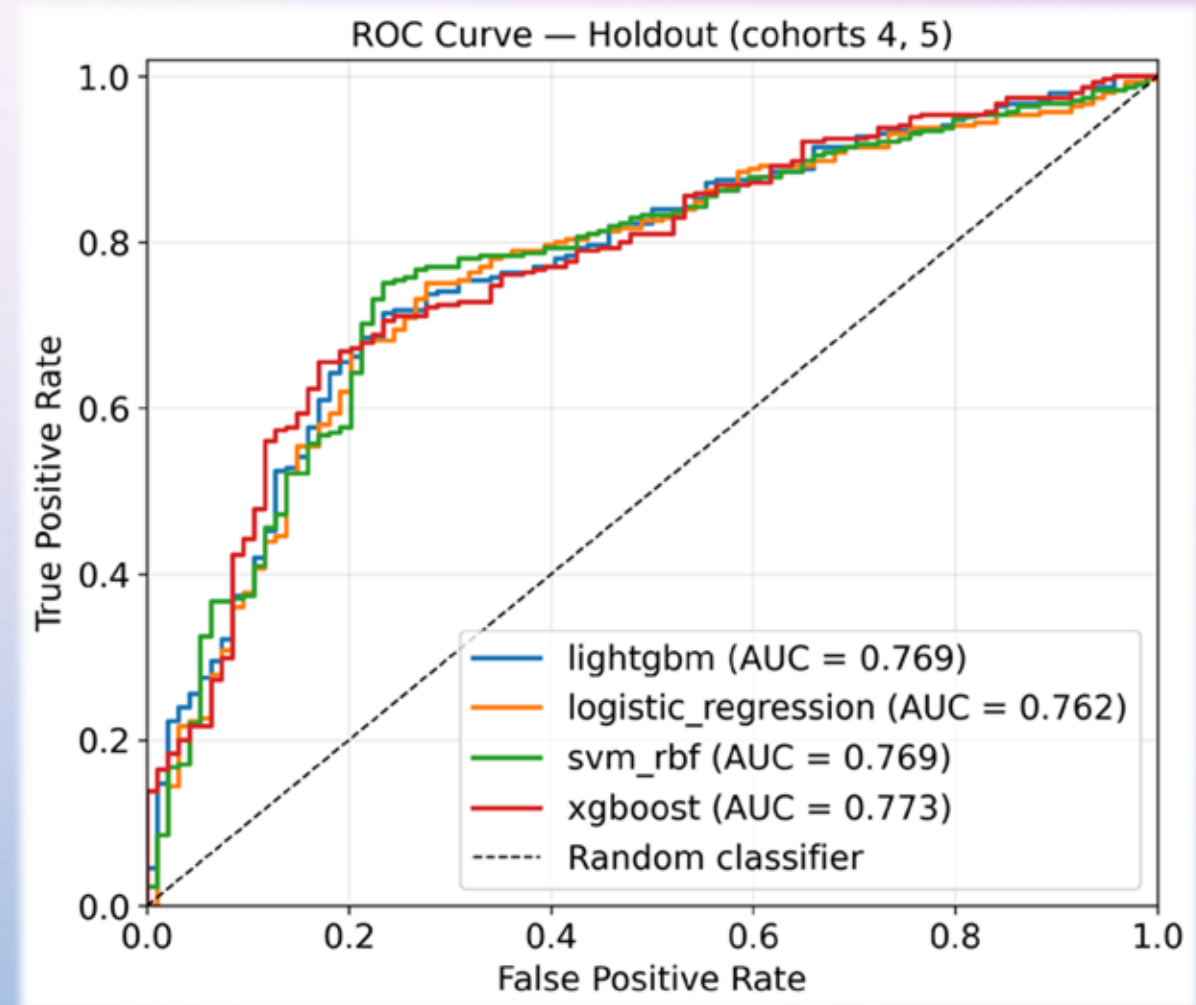
# PREDICTING 5-YEAR SURVIVAL: MODEL COMPARISON

- **Cross-validation** : 1,518 patients (cohorts 1–3), **Test** : 399 patients (cohorts 4–5), **total** : **1,917**
- **Best ROC-AUC:** SVM (0.79).
- **Best minority-class performance:** XGBoost — highest MCC (0.40) and highest F1 on the non-survival ('No') class (0.56).
- **Overall:** all four models cluster tightly (ROC-AUC 0.76–0.79) — no single model dominates across every metric.

|                     | ROC-AUC | PR-AUC | MCC  | Precision survival |      | Recall survival |      | f1 survival |      |
|---------------------|---------|--------|------|--------------------|------|-----------------|------|-------------|------|
|                     |         |        |      | No                 | Yes  | No              | Yes  | No          | Yes  |
| Logistic Regression | 0.76    | 0.89   | 0.36 | 0.39               | 0.90 | 0.78            | 0.62 | 0.53        | 0.74 |
| Light GBM           | 0.76    | 0.91   | 0.39 | 0.41               | 0.90 | 0.74            | 0.67 | 0.55        | 0.78 |
| XGBoost             | 0.77    | 0.91   | 0.40 | 0.43               | 0.91 | 0.78            | 0.69 | 0.56        | 0.78 |
| SVM                 | 0.79    | 0.92   | 0.38 | 0.42               | 0.91 | 0.79            | 0.66 | 0.54        | 0.76 |

# MODEL DISCRIMINATION: ROC CURVE

- **Ranking (holdout AUC):** XGBoost (0.773)  $\approx$  SVM (0.769)  $\approx$  LightGBM (0.769)  $>$  Logistic Regression (0.762) — all four are close.
- **Shape:** All four sit well above the random baseline, with the most separation at low false-positive rates (FPR  $<$  0.2).



# CONCLUSION & FUTURE DIRECTIONS

## Project strengths

- Per-cohort mice imputation preserved all 1,917 patients without leaking data across the holdout split.
- Benchmarked four model families under the same leave-one-cohort-out validation for a fair comparison.
- Feature importance - the model relies on the same factors doctors already trust (age, tumor size, prognostic index) — but subtype and genomic data added real predictive value on top

## Limitations & next steps

- Excluded censored patients (alive, <5yr follow-up) — a Cox/DeepSurv model would use them instead.
- Validated only within METABRIC — next step is an independent cohort like TCGA-BRCA.Clinical + copy-number features only — gene-expression signatures could add more.
- Decision threshold not yet tuned to the real cost of false negatives vs. false positives.